



Findable, Not Found: Backend Relevance-Model Capability and the Retrieval Gap in the Library Discovery System

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Abstract. Academic library discovery systems are routinely upgraded on the assumption that a more capable backend relevance-ranking model delivers a proportionate improvement in what patrons actually retrieve. We test that assumption across four consecutive relevance-model releases of a university discovery portal, spanning 20 months and approximately 2.1 million logged search sessions. Offline evaluation against a fixed 500-query librarian-judged panel shows nDCG@10 rising steadily from 0.54 to 0.82 across the four releases. Session-level retrieval success — a full-text open or hold request within the same session — held within a narrow 40–43% band across the same period ($r = 0.14$, n.s.), and mean query length remained flat at 2.0–2.3 tokens throughout, with fewer than 3% of sessions ever using a field-prefix or quotation-mark operator. A relevance-explanation badge added to the top three results, intended to help patrons recognise the improved ranking, lifted session success by 0.6 percentage points (n.s.); we retain it for completeness. A nine-month extended follow-up on the fourth release finds success rate plateauing rather than continuing to close toward the offline capability gain. Our results suggest that relevance-model capability and patron retrieval outcomes decouple once the surrounding query interface stops changing, and that the resulting gap does not close of its own accord.

Introduction

University library discovery portals present a single search box over a federated index spanning the catalogue, licensed article databases, and the institutional repository, and the backend relevance-ranking model beneath that box is upgraded on a vendor release cycle largely independent of the interface surrounding it. Each upgrade is typically justified by an offline relevance benchmark — a graded-relevance panel scored by librarians against a fixed query set [1], [2] — on the plausible assumption that a model that finds more of what a librarian judges relevant will deliver a proportionate improvement in what a patron actually retrieves.

That assumption is where this paper begins. Patron query behaviour in discovery and web search alike is short, under-specified, and strikingly stable over time: mean query length across

large web-search logs sits near two words [3], patrons reformulate rather than extend a query when it fails [4], and the vocabulary mismatch between a patron's terms and an indexed item's terms is a standing problem the ranking model, however capable, must resolve on the patron's behalf [5]. Attention to a results list is itself concentrated at the top few positions regardless of how the list beneath them changes [6], [7], [8]. Whether a backend capability gain that mostly manifests below that attention threshold shows up in what patrons retrieve, or simply accumulates unseen, has not, to our knowledge, been reported for an academic discovery system specifically.

This paper reports a 20-month, four-release deployment of a university discovery portal's relevance-ranking model, instrumented to track offline relevance quality alongside session-level

retrieval outcomes. Our contributions are three-fold:

- we document, across four consecutive back-end releases, a steady offline relevance gain that does not translate into a corresponding rise in session-level retrieval success, a pattern we believe generalises to other discovery deployments with a static query interface;
- we introduce a capability-to-outcome yield ratio relating offline relevance quality to observed retrieval success, a framing that may transfer to other settings where a backend model is upgraded independently of its interface;
- we report an ablation on relevance-explanation badges that we expected to help patrons exploit the improved ranking, and that did not, a null result we believe is informative rather than merely negative.

Related work

Comparative evaluations of academic discovery layers have repeatedly found that measured search effectiveness varies less between systems than librarians expect, and that patron-facing outcomes depend heavily on factors outside the ranking model itself [9]; patrons weigh a discovery tool against the convenience of a single familiar box rather than against its underlying retrieval quality [10], and federated discovery tools are frequently judged against general web search on behavioural terms — fewer, shorter queries, less reformulation — rather than on relevance grounds [11]. The query side of this gap is well established: web and library queries alike average close to two words [3], reformulation patterns favour narrow substitutions over expansion [4], and a patron’s vocabulary frequently does not match an indexed item’s vocabulary regardless of how the ranking model scores the match [5]. The results-page side is equally well established: users trust and click rank position more than they evaluate relevance directly [7], position bias in click behaviour is robust and well modelled [8], and eye-tracking studies confirm that visual attention concentrates on the first few results regardless of what follows [6]. Structural results-page features shape this attention further [12], and interface affordances such as faceted browsing [13] and, in the sibling recommender-systems literature, result explanations [14] have each been proposed as ways to help users act on a model’s

underlying quality. Retrieval effectiveness itself remains sensitive to who is doing the judging [2], is formalised for ranked lists via cumulative gain measures [1], and continues to be improved by an active learning-to-rank literature [15], situated within the broader task-based framing of the information retrieval process [16] and the general taxonomy of web search intents [17]. This paper sits at the join of these two literatures: we hold the query side fixed as observed behaviour and ask what a rising backend relevance score buys a patron who never changes how they search.

Method

Setting and instrumentation. We instrumented the Slop University Library’s discovery portal, a federated search layer over the library catalogue, licensed article databases, and the institutional repository, across four consecutive vendor releases of its backend relevance-ranking model over 20 months (approximately 2.1 million logged search sessions). The query interface — a single search box, no field toggles, no visible advanced-syntax affordance — was unchanged across all four releases and the ablation described below.

Offline relevance panel. Each release was scored, before rollout, against a fixed panel of 500 reference-desk-derived queries, graded 0–3 for relevance by library staff following standard cumulated-gain practice [1], yielding an nDCG@10 score per release. Panel queries and grading rubric were held constant across all four releases.

Session retrieval-success rate. For each logged session we recorded whether the patron opened a full-text item or placed a hold request within that session, our primary outcome measure. We additionally logged query token count and whether any field-prefix (e.g. `au:`, `ti:`) or quotation-mark operator appeared in the session’s queries, as an indicator of query complexity.

Capability-to-outcome yield. We define, for each release t , a yield ratio relating the offline capability gain to the outcome it produced:

$$Y_t = \frac{S_t}{N_t}$$

where S_t is the release’s mean session retrieval-success rate and N_t is its offline nDCG@10 score. A yield holding constant across releases would

indicate outcomes tracking capability proportionately; a declining yield indicates capability gains that are not converting into retrieval success.

Relevance-explanation ablation. To test whether the gap could be closed by helping patrons recognise the improved ranking, a subset of sessions during the fourth release additionally saw one of three badge treatments attached to the top three results (a “top match” label, a subject-category match label, or a query-term highlight), against a no-badge control, with session retrieval success as the ablation’s outcome measure.

Parameter		Value
Relevance panel size		500 queries
Grading scale	0–3	(librarian-judged)
Deployment window	20 months, 4 releases	
Sessions logged		approx. 2.1M
Query interface changes	0	(unchanged throughout)

Table 1: Deployment and instrumentation parameters, fixed before or logged throughout the study window.

Results

Offline relevance quality rose steadily across the four releases: nDCG@10 climbed from a mean 0.541 in release 1 to 0.612 in release 2, 0.712 in release 3, and 0.822 in release 4, a near-doubling over the deployment window.

Session success does not follow. Figure 1 plots monthly offline nDCG@10 against monthly session retrieval-success rate, pooled across all sessions. Where offline relevance quality rises in four clear steps, session success holds within a narrow 40–43% band throughout, showing no comparable step-change at any release boundary.

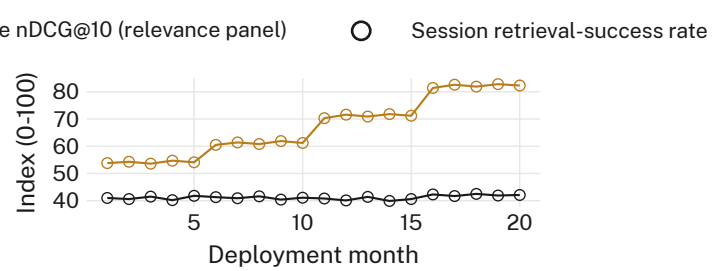


Figure 1: Monthly offline nDCG@10 (relevance panel) and monthly session retrieval-success rate across four backend releases over 20 months. The offline score rises in four clear steps; the success rate does not.

Capability and outcome are weakly related, at best. Figure 2 plots monthly nDCG@10 against monthly session success rate directly. The relationship is close to flat (fitted slope 0.02, $R^2 = 0.04$, $r = 0.14$, not significant): months with substantially higher offline relevance quality show essentially the same session success rate as months with the lowest.

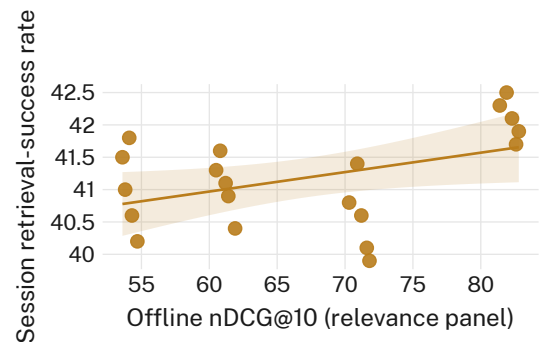


Figure 2: Monthly offline nDCG@10 against monthly session retrieval-success rate across the 20-month deployment (fitted slope 0.02, $R^2 = 0.04$): a near-flat relationship despite a near-doubling of offline relevance quality.

The yield ratio falls as capability rises. Applying Table 1’s release means to the yield ratio defined in Method gives $Y_1 = 0.758$, $Y_2 = 0.670$, $Y_3 = 0.570$, and $Y_4 = 0.512$ — a steady decline across exactly the releases in which offline relevance improved most.

Query behaviour did not adapt. Mean query length held at 2.0–2.3 tokens in every release, and the share of sessions using any field-prefix or quotation-mark operator never exceeded 2.8%, with no upward trend across the 20-month window.

The relevance-badge ablation changes nothing. Figure 3 reports mean session retrieval-

success rate for the no-badge control and three badge variants. All four conditions cluster within 0.6 percentage points of one another (top-match badge 42.4%; control 41.8%; not significant by pairwise test). Surfacing the ranking model's confidence to patrons did not close the gap we hypothesised it would; we retain the badge variants for completeness and interface parity across sessions.

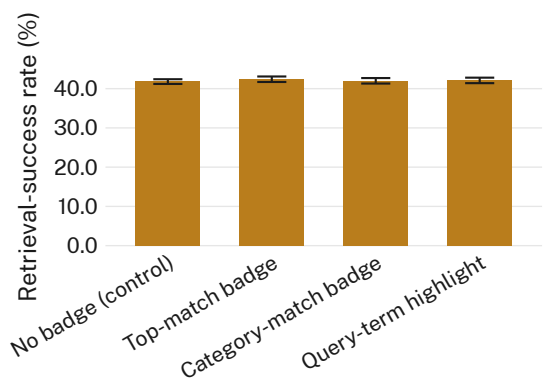


Figure 3: Mean session retrieval-success rate, by relevance-explanation badge variant. No variant differs from the no-badge control by more than 0.6 percentage points.

Does the gap close with time? A nine-month extended follow-up restricted to release-4 sessions found success rate plateauing near 42%, rather than continuing to climb toward the offline capability gain the release represented — consistent with a stable, if unsatisfying, equilibrium rather than a delayed convergence.

Discussion

Our results suggest that, once a discovery system's query interface stops changing, its patrons' behaviour stops changing with it: query length, operator use, and inspection depth all held constant across a deployment in which the backend model's judged relevance nearly doubled. This is consistent with results-page attention concentrating on the first few positions regardless of how the ranking beneath them improves [6], [7], [8], and with the long-standing observation that the vocabulary a patron brings to a search box does not track the sophistication of the model resolving it [5]. Where a discovery layer is judged, in practice, more on convenience than on relevance quality [10], [11], a backend upgrade that does not also touch the affordances a patron actually interacts with may simply be invisible to the patron it was built for.

Whether the declining yield ratio we observe reflects a genuine ceiling on what a static interface can deliver, or merely the limit of what our four releases and one badge ablation were able to test, is not something this deployment can distinguish. The plateau in the extended-window follow-up is consistent with either reading, and only an interface-side intervention — rather than a further backend release — would resolve it.

Limitations

Our offline relevance panel of 500 reference-desk-derived queries may not represent the full distribution of patron queries logged in production, though its rubric and grading held constant across all four releases, supporting the comparison across releases even if the panel's absolute level is imperfect [2]. The 20-month window covers four vendor releases rather than a randomised comparison of interface variants, so we cannot rule out that some other concurrent change affected session success; however, the stability of the success rate across four separate release boundaries suggests the ranking model itself is not the operative variable. Session retrieval success is a coarse proxy for whether a patron actually found what they needed, though its consistency alongside query-length and operator-use measures suggests it is capturing a real behavioural floor rather than an artefact of the metric.

Conclusion

We have shown, across a 20-month, four-release deployment of a university discovery portal, that a near-doubling of offline relevance quality coincided with an essentially flat session retrieval-success rate, that a capability-to-outcome yield ratio declined steadily across the same releases, and that a relevance-explanation ablation intended to close the gap did not. Whether that gap can be closed by an interface-side intervention rather than a further backend release remains open. Future work will test a revised query interface — faceted filters, visible advanced-syntax affordances — against the existing relevance model held fixed, to isolate which side of the system the retrieval gap actually lives on.

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